

# Adam Jarick

SOFTWARE ENGINEER · FULL STACK & AI/ML

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## Summary

Full stack engineer with two years building AI and data products end to end, from PyTorch models and LLM pipelines through to the React front ends and Kubernetes infrastructure that serve them. At Building Commission NSW I own entity resolution across hundreds of millions of records and the fuzzy search that runs on top of it. Before that I was one of the first engineers at a scam intelligence startup that grew from zero to over \$1M in annual revenue, where I was sole architect of the conversational agents that harvested hundreds of thousands of intelligence artefacts.

## Skills

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|---------------------------|----------------------------------------------------------------------------------------|
| <b>Languages</b>          | Python, TypeScript, JavaScript, Go, SQL, Java                                          |
| <b>AI &amp; ML</b>        | PyTorch, OpenAI API, ONNX Runtime, Neural network design, LLM integration              |
| <b>Backend</b>            | Node.js, Hono, Express, Flask, Go Chi, REST APIs, WebSockets, OAuth, JWT, Keycloak     |
| <b>Frontend</b>           | React, Next.js, Angular, Tailwind, Electron, HTML, CSS                                 |
| <b>Databases</b>          | PostgreSQL, MS SQL Server, ClickHouse, Elasticsearch, Redis, MongoDB                   |
| <b>Data Platform</b>      | Databricks, Azure Data Factory, Splink, Airbyte, Superset, Prisma                      |
| <b>Testing</b>            | Unit and integration testing, Jest, Playwright, A/B testing                            |
| <b>Cloud &amp; DevOps</b> | Azure, AWS, GCP, Docker, Kubernetes, Helm, GitHub Actions, ArgoCD, Cloudflare, Grafana |

## Work Experience

### Building Commission NSW, Department of Customer Service

Sydney, NSW

Software Engineer

Apr 2026 - Present

- Own the entity resolution platform that consolidates hundreds of millions of names and companies from close to 100 source systems into deduplicated, mutually enriched master tables, built on Splink probabilistic record linkage in Databricks.
- Built and own the Elasticsearch layer that gives analysts and building inspectors fuzzy name and company search across those hundreds of millions of rows.
- Contributed to the risk rating that ranks buildings for inspection, directing a limited inspector workforce towards the sites most likely to be non-compliant.
- Helped build the alerting system that sends HTML and CSS email notifications to the data analytics team and field inspectors, delivered through the Postal mail server over SMTP.
- Develop ETL pipelines in Azure Data Factory and Databricks that feed the master tables from Microsoft SQL Server and legacy sources.
- Sole full stack engineer in a team of 15+ otherwise specialised in backend and data engineering, so I own all front end work, and the point of contact with the IBM i2 and iBase team for maintenance and integration.

### Apate.AI

Sydney, NSW

Software Engineer

Oct 2024 - Apr 2026

- Joined early as one of four engineers and helped grow the platform from zero to over \$1M in annual revenue.
- Cut voice bot response latency from 1.2 seconds to 150 milliseconds by standing up an Australian ElevenLabs endpoint on Cloudflare edge infrastructure.
- Sole architect and developer of the anti-scam conversational agents for WhatsApp, Telegram and email, designed and built end to end in TypeScript with WebSocket protocol manipulation and OpenAI models to engage scammers and extract intelligence.
- Those agents harvested hundreds of thousands of intelligence artefacts for clients, including bank accounts, cryptocurrency wallets, phone numbers, addresses and names.
- Built the REST backend on Hono and Prisma over PostgreSQL, covering chat management, bot configuration and real-time state monitoring, covered by unit and integration tests.
- Architected and built the client insights portal on my own in React and TypeScript over ClickHouse, Redis, Airbyte and Superset, serving hundreds of users authenticated through Keycloak and JWT.
- Containerised and deployed the platform to Kubernetes on Azure, AWS and GCP using Helm charts, GitHub Actions pipelines and monitoring infrastructure.
- Onboarded and supported 10 enterprise clients in a forward deployed capacity, and owned one major account end to end, running requirements sessions and translating business KPIs into engineering work.

### Webschool.au

Macquarie Park, NSW

AI Engineer, Internship

Jun 2025 - Nov 2025

- Built a voice authentication system for platform login as a second authentication factor, using a Siamese neural network in PyTorch trained on VoxCeleb.
- Extracted 13 Mel-frequency cepstral coefficients per audio frame and scored speaker pairs by cosine similarity to verify identity.
- Containerised the model and deployed it into the company's existing Azure portal infrastructure.

# Security Research

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## Bug Bounty, Bugcrowd & Intigriti

Independent Researcher | [bugcrowd.com/h/Adstar](https://bugcrowd.com/h/Adstar)

2026 - Present

- 11 accepted vulnerability submissions across Bugcrowd and Intigriti programs, including one rated P1, the highest severity assigned.
- The P1 was an unauthenticated account takeover on bpost's MyPreferences delivery API: the backend trusted a client-supplied identity header, so any customer's profile could be read, modified or deleted with only their email address and a public API key (CVSS 9.8).

## Projects

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### EquityLens: Sector-Calibrated ASX Equity Screener

[equitylens.money](https://equitylens.money) | [github.com/Adstar123/EquityLens](https://github.com/Adstar123/EquityLens)

Mar 2026 - Aug 2026

- Built and deployed the full stack Go and Angular platform end to end, scoring every company in the ASX 300 against configurable financial models across 11 sectors and 7 weighted ratios.
- Wrote the scoring engine as a pure function with outlier clamping, weight redistribution when ratios are unavailable and a missing-data threshold, covered by unit tests.
- Built a superadmin interface so a finance collaborator, who owns the financial modelling, defines and tunes the ratios, weights, benchmarks and thresholds himself, with preview, publish and version history and no code changes or redeployment.
- Ingest and normalise fundamentals and prices from Yahoo Finance, Alpha Vantage and MarketIndex into PostgreSQL behind a Redis cache, on GitHub Actions cron schedules aligned to ASX trading hours.
- Front end is an Angular 19 SPA using signals and standalone components with Tailwind and ECharts, authenticated by OAuth2 PKCE through Google and GitHub with JWT sessions.

### OpenGTO: Neural Network Poker Preflop Trainer

[github.com/Adstar123/OpenGTO](https://github.com/Adstar123/OpenGTO)

May 2025 - Dec 2025

- Built a Deep Counterfactual Regret Minimisation system in PyTorch where three neural networks converge towards Nash equilibrium through self-play across 20,000+ training iterations.
- Engineered a 317-dimensional feature extraction pipeline and an 8-stage curriculum learning system with weighted reservoir sampling over 5 million+ training samples.
- Shipped it as a cross-platform Electron, React and TypeScript desktop application with interactive table visualisation, real-time feedback and a range viewer covering all 169 starting hands.

### Portfolio Website

[adamjarick.com](https://adamjarick.com)

Feb 2026 - Present

- Built in Next.js, React 19, TypeScript, Tailwind and Three.js, running the OpenGTO model directly in the browser through ONNX Runtime.
- Deployed on Vercel with Content Security Policy and strict transport security headers, plus server-side rate limiting and input sanitisation on the contact form.

### Chrome Extension AI Co-Pilot

Honours Thesis, Macquarie University

Feb 2024 - Nov 2024

- Built a Chrome extension that read student learning analytics by injecting JavaScript into iframes to reach SCORM and Moodle APIs, then returned AI-driven recommendations and feedback.
- Trial results showed increases in student engagement, results and motivation to keep learning.

## Education

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### Macquarie University

North Ryde, NSW

Double degree: Software Engineering (BE Honours) and Cyber Security (BInfoTech)

Jan 2021 - Nov 2025

Graduated with a Distinction average. Holds a current Baseline security clearance.